



Instructions for Exchange of Sub-assemblies

Gas Engine

8V4000L32F

12V4000L32F

16V4000L32F

20V4000L32F

8V4000L33F

12V4000L33F

16V4000L33F

20V4000L33F

MS22050/02E

Engine model	kW/cyl.	Application group
8V4000L32F	100 kW/cyl.	3A, continuous operation, unrestricted
12V4000L32F	100 kW/cyl.	3A, continuous operation, unrestricted
16V4000L32F	100 kW/cyl.	3A, continuous operation, unrestricted
20V4000L32F	100 kW/cyl.	3A, continuous operation, unrestricted
8V4000L33F	110 kW/cyl.	3A, continuous operation, unrestricted
12V4000L33F	110 kW/cyl.	3A, continuous operation, unrestricted
16V4000L33F	110 kW/cyl.	3A, continuous operation, unrestricted
20V4000L33F	110 kW/cyl.	3A, continuous operation, unrestricted

Table 1: Applicability

Table of Contents

1	Preface			
1.1	Preface	6		
2	Safety			
2.1	Important provisions for all products	7		
2.2	Intended use of all products	9		
2.3	Personnel and organizational requirements	10		
2.4	Safety regulations for initial start-up and operation	11		
2.5	Safety regulations for commissioning and operation, specific information for gaseous fuel supply or and dual fuel applications	13		
2.6	Safety regulations for assembly, maintenance, and repair work	15		
2.7	Safety regulations for assembly, maintenance and servicing work, specific information for gaseous fuel and dual fuel supply or applications	19		
2.8	Fire and environmental protection, indirect materials	20		
2.9	Fire prevention and environmental protection, fluids and lubricants, auxiliary materials, specific information for gaseous fuel supply in gaseous-fuel or dual-fuel applications	23		
2.10	Standards for warning notices in the text and highlighted information	24		
3	General Information			
3.1	Engine side and cylinder designations	25		
4	Technical Data			
4.1	Settings	26		
5	Task Description			
5.1	Crankcase and Attachments	28		
5.1.1	Inspection port cover - Overview	28		
5.1.2	Inspection port cover - Preparatory steps	30		
5.1.3	Inspection port cover - Removal	31		
5.1.4	Inspection port cover sealing surfaces - Cleaning	32		
5.1.5	Inspection port cover - Installation	33		
5.1.6	Inspection port cover - Check	34		
5.1.7	Inspection port cover - Final steps	35		
5.1.8	Cylinder liner - Overview	36		
5.1.9	Cylinder liner - Preparatory steps	38		
5.1.10	Cylinder liner - Removal	39		
5.1.11	Cylinder liner - Cleaning	41		
5.1.12	Cylinder liner - Check	42		
5.1.13	Cylinder liner - Tolerances	43		
5.1.14	Cylinder liner - Installation	45		
5.1.15	Cylinder liner - Final steps	47		
5.2	Running Gear	48		
5.2.1	Crankshaft PTO, free end - Overview	48		
5.2.2	Vibration damper on PTO, free end - Removal	53		
5.2.3	Vibration damper on PTO, free end - Installation	58		
5.2.4	Pistons and conrod - Overview	67		
5.2.5	Pistons and conrods - Preparatory steps	70		
5.2.6	Piston and conrod - Removal	71		
5.2.7	Piston and conrod - Disassembly	73		
5.2.8	Piston - Cleaning	75		
5.2.9	Piston - Check	77		
5.2.10	Piston - Tolerances	79		
5.2.11	Conrod - Check	82		
5.2.12	Conrod - Tolerances	84		
5.2.13	Conrod - Assembly	88		
5.2.14	Piston - Assembly	91		
5.2.15	Piston and conrod - Installation	94		
5.2.16	Piston and conrod - Final steps	100		
5.3	Cylinder Head	101		
5.3.1	Cylinder head with add-on components - Overview	101		
5.3.2	Cylinder head - Preparatory operations	103		
5.3.3	Cylinder head - Removal	104		
5.3.4	Cylinder head - Disassembly	107		
5.3.5	Cylinder head - Cleaning	109		
5.3.6	Cylinder head - Assembly	111		
5.3.7	Cylinder head - Installation	116		
5.3.8	Cylinder head - Final steps	122		
5.4	Valve Gear	123		
5.4.1	Valve gear - Overview	123		
5.4.2	Valve drive - Overview	125		
5.4.3	Valve drive - Preparatory steps	130		
5.4.4	Valve gear - Removal	131		
5.4.5	Valve drive - Cleaning	132		
5.4.6	Valve drive - Check	133		
5.4.7	Valve drive - Installation	134		
5.4.8	Valve drive - Final steps	137		
5.5	Gas System	138		
5.5.1	Gas system - Overview	138		
5.5.2	Gas control valve (TecJet) - Preparatory steps	140		
5.5.3	Gas control valve (TecJet) - Removal	141		
5.5.4	Gas control valve (TecJet) - Check	142		
5.5.5	Gas control valve (TecJet) - Installation	143		
5.5.6	Gas control valve (TecJet) - Final steps	145		
5.6	Exhaust Turbocharger	146		
5.6.1	Exhaust turbocharger - Overview	146		
5.6.2	Exhaust turbocharger - Preparatory steps	150		

5.6.3 Exhaust turbocharger - Removal	151	5.9.7 Exhaust pipework after exhaust turbocharger - Overview	210
5.6.4 Exhaust turbocharger - Cleaning	152	5.9.8 Exhaust pipework after turbocharger - Removal	213
5.6.5 Exhaust turbocharger - Check	153	5.9.9 Exhaust pipework after exhaust turbocharger - Installation	214
5.6.6 Exhaust turbocharger - Installation	154		
5.6.7 Exhaust turbocharger - Final steps	157		
5.7 Mixture Cooling	158	5.10 Starting Equipment	215
5.7.1 Mixture cooler - Overview	158	5.10.1 Starter - Overview	215
5.7.2 Mixture cooler - Preparatory steps	160	5.10.2 Starter - Preparatory steps	217
5.7.3 Mixture cooler - Removal	161	5.10.3 Starter - Removal	218
5.7.4 Mixture cooler - Cleaning	162	5.10.4 Starter - Cleaning	219
5.7.5 Mixture cooler - Check	163	5.10.5 Starter - Check	220
5.7.6 Mixture cooler - Installation	164	5.10.6 Starter - Installation	221
5.7.7 Mixture cooler - Final steps	165	5.10.7 Starter - Final steps	223
5.8 Air Intake / Air Supply	166	5.11 Lube Oil System, Lube Oil Circuit	224
5.8.1 Air intake / mixture supply - Overview	166	5.11.1 Oil heat exchanger - Overview	224
5.8.2 Air filter and air intake - Overview	170	5.11.2 Oil heat exchanger - Preparatory steps	226
5.8.3 Air filter and air intake - Removal	171	5.11.3 Oil heat exchanger - Removal	227
5.8.4 Air filter and air intake - Installation	172	5.11.4 Oil heat exchanger - Cleaning	228
5.8.5 Mixture line before mixture cooler - Overview	173	5.11.5 Oil heat exchanger - Installation	229
5.8.6 Mixture pipe upstream of mixture cooler - Preparatory steps	177	5.11.6 Oil heat exchanger - Final steps	230
5.8.7 Mixture line upstream of mixture cooler - Removal	178	5.11.7 Exhaust turbocharger oil supply, 16V - Overview	231
5.8.8 Mixture line before mixture cooler - Installation	179	5.11.8 Exhaust turbocharger oil supply, 16V - Removal	232
5.8.9 Mixture pipe upstream of mixture cooler - Final steps	183	5.11.9 Exhaust turbocharger oil supply, 16V - Installation	233
5.8.10 Mixture line downstream of mixture cooler - Overview	184	5.12 Coolant System	234
5.8.11 Mixture pipe downstream of mixture cooler - Preparatory steps	187	5.12.1 Coolant pipework, HT circuit - Overview	234
5.8.12 Mixture pipework downstream of mixture cooler - Removal	188	5.12.2 Coolant pipework, HT circuit - Preparatory steps	235
5.8.13 Mixture line after mixture cooler - Cleaning	189	5.12.3 Coolant pipework, HT circuit - Removal	236
5.8.14 Mixture line downstream of mixture cooler - Installation	190	5.12.4 Coolant pipework, HT circuit - Installation	237
5.8.15 Mixture pipe downstream of mixture cooler - Final steps	191	5.12.5 Coolant pipework, HT circuit - Final steps	239
5.8.16 Heat shield for mixture line - Overview	192	5.12.6 Coolant pipework from/to mixture cooler (HT- circuit) - Overview	240
5.8.17 Jacket for mixture line - Removal	193	5.12.7 Coolant pipework from/to mixture cooler (HT system) - Preparatory steps	242
5.8.18 Heat shield for mixture line - Installation	194	5.12.8 Coolant pipework from/to mixture cooler (HT circuit) - Removal	243
5.8.19 Bypass line - Overview	195	5.12.9 Coolant pipework from/to mixture cooler (HT circuit) - Installation	244
5.8.20 Bypass line - Removal	197	5.12.10 Coolant pipework from/to mixture cooler (HT system) - Final steps	245
5.8.21 Bypass line - Installation	198	5.12.11 Coolant ventilation - Overview	246
5.9 Exhaust Pipework	199	5.12.12 Coolant ventilation - Preparatory steps	250
5.9.1 Exhaust pipework - Overview	199	5.12.13 Coolant ventilation - Removal	251
5.9.2 Exhaust pipework downstream of cylinder head - Overview	203	5.12.14 Coolant ventilation - Installation	252
5.9.3 Exhaust pipework downstream of cylinder head - Preparatory steps	206	5.12.15 Coolant ventilation - Final steps	253
5.9.4 Exhaust pipework after cylinder head - Removal	207	5.12.16 Coolant pipework from/to exhaust turbochargers - Overview	254
5.9.5 Exhaust pipework after cylinder head - Installation	208	5.12.17 Coolant pipework from/to exhaust turbochargers - Preparatory steps	257
5.9.6 Exhaust pipework after cylinder head - Final steps	209	5.12.18 Coolant pipework from/to exhaust turbochargers - Removal	258

5.12.19	Coolant pipework from/to exhaust turbochargers - Installation	259	5.13.10	Engine Control Unit ECU, Engine Monitoring Unit EMU and knock module - Installation	306
5.12.20	Coolant pipework from/to exhaust turbochargers - Final steps	260			
5.13	Monitoring, Control and Regulation Equipment	261	6	Appendix A	
5.13.1	Engine wiring - Overview	261	6.1	Tightening specifications for screws, nuts and bolts	307
5.13.2	Sensors - Overview	269	6.2	Conversion tables	320
5.13.3	Sensors - Preparatory steps	281	6.3	Tolerances and wear limits list - General information	324
5.13.4	Sensors - Removal	282	6.4	Abbreviations	325
5.13.5	Sensors - Check	289	6.5	Contact person/Service partner	327
5.13.6	Sensors - Installation	290			
5.13.7	Sensors - Final steps	300	7	Appendix B	
5.13.8	Engine Control Unit ECU, Engine Monitoring Unit EMU, knock module - Overview	301	7.1	Special Tools	328
5.13.9	Engine Control Unit ECU, Engine Monitoring Unit EMU and knock module - Removal	305	7.2	Index	346

1 Preface

1.1 Preface

This manual provides instructions for your product from the manufacturer MTU.

Definition of MTU

MTU refers to Rolls-Royce Power Systems AG and MTU Friedrichshafen GmbH or an affiliated company pursuant to Section § 15 AktG (German Stock Corporation Act) or a controlled company (joint venture).

2 Safety

2.1 Important provisions for all products

General

This product may pose a risk of injury or damage in the following cases:

- Incorrect use
- Operation, maintenance and repair by unqualified personnel
- Changes or modifications which are neither made nor authorized by the manufacturer
- Noncompliance with the safety instructions and warning notices

Nameplates

The product is identified by nameplate, model designation or serial number. This data must match the specifications in these instructions.

Nameplates, model designation or serial number can be found on the product.

All EU-certified engines delivered by MTU come with a second nameplate. This second nameplate is delivered "loosely" with the engine. If the nameplate secured to the engine after installation in the vehicle/system is not visible without the removal of components, the system integrator must install the second nameplate in a clearly visible area on the vehicle/system.

Emissions regulations and emissions label

Responsibility for compliance with emissions regulations

Modification or removal of any mechanical/electronic components or the installation of additional components including the execution of calibration processes that might affect the emissions characteristics of the product are prohibited by emissions regulations. Emissions-related components must only be serviced, exchanged or repaired if the components used for this purpose are approved by the manufacturer.

Noncompliance with these specifications will invalidate the design type approval or certification issued by the emissions regulation authorities. The manufacturer does not accept any liability for violations of the emissions regulations.

The product must be operated over its entire life cycle according to the conditions defined as "Intended use" (→ Page 9).

Emissions certification applicable to engines with EPA Nonroad Tier 4 emissions certification in accordance with 40 CFR 1039 and with EPA Locomotive emissions certification in accordance with 40 CFR 1033

Extract from the standard:

Failing to follow these instructions when installing a certified engine in a piece of nonroad equipment violates federal law (40 CFR 1068.105(b)), subject to fines or other penalties as described in the Clean Air Act.

Extract from the standard:

If you install the engine in a way that makes the engine's emission control information label hard to read during normal engine maintenance, you must place a duplicate label on the equipment, as described in 40 CFR 1068.105.

When fitting the second label, the requirements of 40 CFR 1068.105(c) must be followed and observed. This paragraph describes the process for requesting and fitting the label, the documentation obligations and storage obligations for the required documents.

Replacing components with emissions labels

On all engines from MTU fitted with emissions labels, these labels must remain on the engine throughout its operational life.

Exception: Engines used exclusively in land-based, military applications other than by US government agencies.

Please note the following when replacing components with emissions labels:

- The relevant emissions labels must be affixed to the spare part.
- Emissions labels shall not be transferred from the replaced part to the spare part.
- The emissions labels must be removed from the replaced part and destroyed.

2.2 Intended use of all products

Intended use

The product is intended for use in accordance with its contractually-defined purpose as described in the relevant technical documents only.

Intended use entails operation:

- Within the permissible operating parameters in accordance with the (→ Technical data)
- With fluids and lubricants approved by the manufacturer in accordance with the (→ Fluids and Lubricants Specifications of the manufacturer)
- With preservation approved by the manufacturer in accordance with the (→ Preservation and Represervation Specifications of the manufacturer)
- With spare parts approved by the manufacturer in accordance with the (→ Spare Parts Catalog/MTU contact/Service partner)
- In the original as-delivered configuration or in a configuration approved by the manufacturer in writing (also applies to engine control/parameters)
- In compliance with all safety regulations and in adherence with all safety and warning notices in this manual
- In compliance with the maintenance work and intervals specified in the (→ Maintenance Schedule) throughout the useful life of the product
- Work which requires communication with the control units must only be carried out with a diagnosis software or with tools approved by the manufacturer.
- In compliance with the maintenance and repair instructions contained in this manual, in particular with regard to the specified tightening torques
- With the exclusive use of technical personnel trained in commissioning, operation, maintenance and repair

The product must not be operated in explosive atmospheres unless the engine fulfills the conditions for such use and approval has been granted.

Any other use, particularly misuse, is considered as being contrary to the intended purpose. Such improper use increases the risk of injury and damage when working with the product. The manufacturer shall not be held liable for any damage resulting from improper, non-intended use.

The specifications of the manufacturer will be amended or supplemented as necessary. Prior to operation, make sure that the latest version is used. The latest version can be found at:

- <http://www.mtu-solutions.com>

Modifications or conversions

Unauthorized changes to the product represent a contravention of its intended use and compromise safety.

Changes or modifications shall only be considered to comply with the intended use when expressly authorized by the manufacturer. The manufacturer shall not be held liable for any damage resulting from unauthorized changes or modifications.

2.3 Personnel and organizational requirements

Organizational measures of the user/manufacturer

This manual must be issued to all personnel involved in operation, maintenance, repair, assembly, installation, or transportation.

Keep this manual handy in the vicinity of the product such that it is accessible to operating, maintenance, repair, assembly, installation, and transport personnel at all times.

Personnel must receive instruction on product handling and repair based on this manual. In particular, personnel must have read and understood the safety requirements and warnings before starting work.

This is important in the case of personnel who only occasionally perform work on or around the product. Such personnel must be instructed repeatedly.

Personnel requirements

All work on the product must be carried out by trained, instructed and qualified personnel only:

- Training at the Training Center of the manufacturer
- Qualified personnel from the areas mechanical engineering, plant construction, and electrical engineering and also for work with live parts

The operator must define the responsibilities of the personnel involved in operation, maintenance, repair, assembly, installation, and transport in writing.

Personnel shall not report for duty under the influence of alcohol, drugs or strong medication.

Clothing and personal protective equipment

Always wear appropriate personal protective equipment, e.g. safety shoes, ear protectors, protective gloves, goggles, breathing mask. Follow the instructions concerning personal protective equipment in the respective descriptions of the individual activities or the manufacturer's documentation included in the delivery.

When working on live components, in particular on electrical switchgear or on high-voltage component, wear special protective clothing according to IEC 61482.

Safe handling of Substances of Very High Concern pursuant to the REACH regulation (Registration, Evaluation, Authorization and restriction of CHemicals): We recommend wearing protective gloves at all times in order to reduce risk when working.